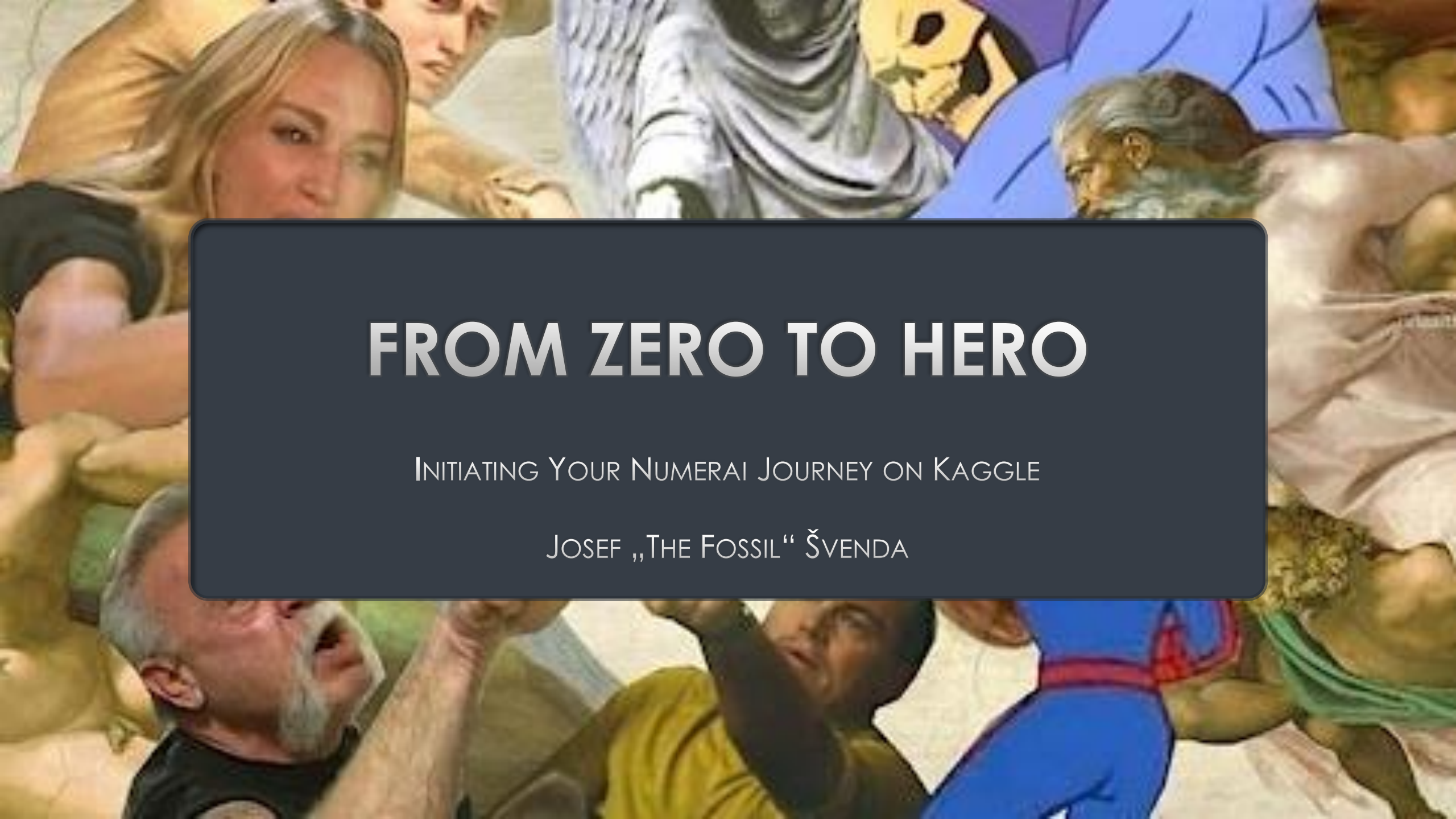




FROM ZERO TO HERO

INITIATING YOUR NUMERAI JOURNEY ON KAGGLE

JOSEF „THE FOSSIL“ ŠVENDA

AGENDA

WHAT IS KAGGLE?

NUMERAI DATA

NUMERAI EXAMPLE NOTEBOOKS

NUMERAI AUTOMATION TIPS&TRICKS



WHAT IS KAGGLE?

kaggle is a data science competition platform and online community of data scientists and machine learning practitioners under Google LLC.

As of October 2023, it has over 15 million users in 194 countries.

The screenshot shows the Kaggle user dashboard for Josef Švenda. The interface includes a left sidebar with navigation links, a top search bar, and a main content area with a welcome message and various statistics.

Left Sidebar:

- Create
- Home
- Competitions
- Datasets
- Models
- Code
- Discussions
- Learn
- More
 - User Rankings
 - Blog
 - Documentation
 - Progression
 - Host a Competition
 - KaggleX Mentorship
 - Support/Contact
 - Community Guidelines
 - View Active Events

Top Bar:

- Search
- LOGIN STREAK
- TIER PROGRESS
- PUBLIC ACTIVITY

Welcome, Josef Švenda!

Jump back in, or start something new.

Statistics:

- 47** Your longest is 181 days
- 60%** to Expert
- MTWTFSS** (activity calendar)

Activity Summary:

- Notebooks:** 216 total created (up 2)
- Discussions:** 76 total posted (up 5)
- Competitions:** 11 total joined
- Datasets:** 9 total created
- Courses:** 12 total completed

How to start: Choose a focus for today

Help us make relevant suggestions for you

Footer:

- Competitions
- Learn
- Discussions

LIMITS OF FREE

WHAT YOU CAN DO IN KAGGLE

Limits

4 CPUs, 30GB RAM

GPU, 4 CPUs, 29 GB
RAM, weekly max. 30 h

TPU, 96 CPUs, 330 GB
RAM, weekly max. 20 h

Single run max. 12 hours
CPU&GPU, 9h TPU

20GB disk space

Max 5 ntbks concurrently
running

Chaining of notebooks

Datasets

Kaggle API
running Kaggle
notebooks from outside

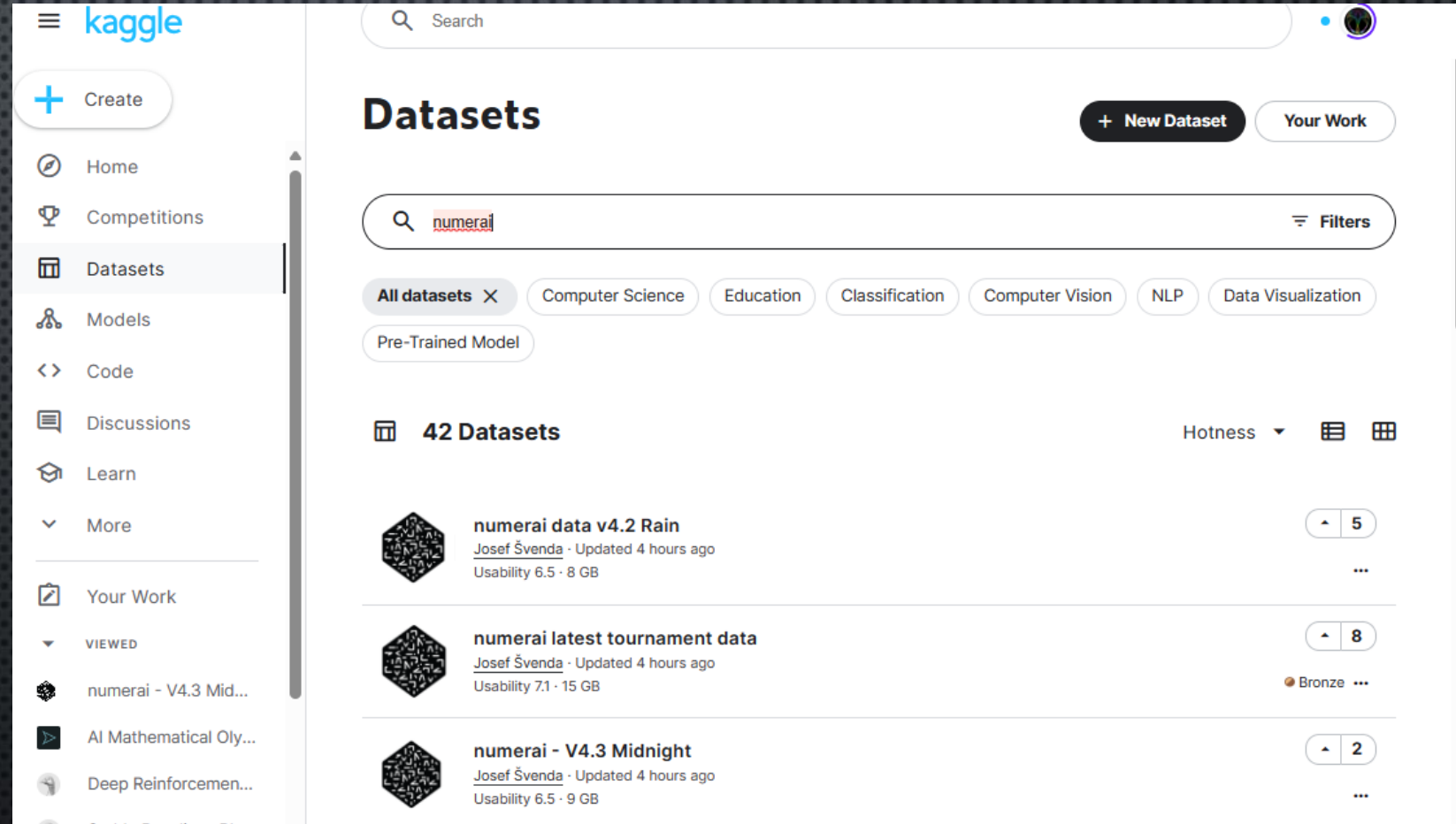
Enablers



NUMERAI DATASETS

Tournament data are weekly downloaded and stored as public Kaggle datasets:

- V4.3 Midnight
- V4.2 Rain
- V4 & V4.1



The screenshot shows the Kaggle website's 'Datasets' section. A search bar at the top contains the text 'numeraid'. Below the search bar, there are filters for 'All datasets', 'Computer Science', 'Education', 'Classification', 'Computer Vision', 'NLP', and 'Data Visualization'. The search results show 42 datasets. The first three results are:

- numeraid data v4.2 Rain** by Josef Švenda, updated 4 hours ago, with a usability of 6.5 and a size of 8 GB. It has 5 votes.
- numeraid latest tournament data** by Josef Švenda, updated 4 hours ago, with a usability of 7.1 and a size of 15 GB. It has 8 votes and a Bronze medal.
- numeraid - V4.3 Midnight** by Josef Švenda, updated 4 hours ago, with a usability of 6.5 and a size of 9 GB. It has 2 votes.

NUMERAI DATASETS

Each dataset has the producing notebook doing:

- Downloading current data
- „EDA“
- Producing smaller data files with „non-overlapping“ eras

Either the dataset or the producing notebook can be used as input for other notebooks.

The screenshot shows the Kaggle interface for the 'numeraidata v4.3 Midnight' dataset. The left sidebar contains navigation links: Home, Competitions, Datasets, Models, Code (highlighted), Discussions, Learn, and More. Below these is a 'Your Work' section with a 'VIEWED' list containing 'numeraidata - V4.3 Mid...', 'AI Mathematical Oly...', 'Deep Reinforcemen...', and 'Stable Baselines RL'. The main content area features a search bar and the dataset title 'numeraidata v4.3 Midnight'. Below the title are tabs for Notebook, Input, Output, Logs, Comments (0), and Settings. The 'Notebook' tab is active, showing a notebook titled 'Kaggle notebook with latest Numerai v4.3 data'. The notebook text explains that it automatically downloads the latest Numerai v4.3 data (a.k.a. Midnight) each Saturday. It mentions that older datasets (V4 and V4.1) are downloaded by the 'numeraidata' notebook, while V4.2 is downloaded by the 'numeraidata v4.2 Rain' notebook. The notebook also states that the output directories will contain the latest download of data in parquet format, which can be used for notebook chaining and EDA. A note mentions that up to 20 GBs of output from a Notebook may be saved to disk in /kaggle/working, and that this data is saved automatically and can be reused in future Notebooks. A 'Table of Contents' on the right lists the notebook's sections: 'Kaggle notebook with latest Numer...', 'Libraries and parameters', 'Downloading current era dataset', 'Info about dataset', and 'Separating integer datasets into fil...'. The top right of the page shows a user profile icon and a '4' in a circle, indicating the number of versions or iterations.

There are 5 symbolically staked uploaded public models based on [Numerai tutorials](#):

- | | | | 1 NMR ≈ \$24.88 | Tournaments ▾ | Hedge Fund | Discord | GitHub | Forum | Docs | | | |
|--------|-----|----------------------|------------------------|---------------------------------|-------------------------|---------|---------|--------|-----------|-------|-------|----|
| | | | Data Scientists | Models | | | | | | | | |
| | | | Staked Models
6,853 | Total NMR Staked
637,695 NMR | Avg 1Y Return
34.45% | | | | | | | |
| KAGGLE | | | CORJ60 | FNCV3 | CORR20V2 | MMC | BMC | CORT20 | STAKE | 1D | ▼ 3M | 1Y |
| 128 | agg | JOS_KAGGLE_HELLO | 0.0110 | 0.0035 | 0.0046 | -0.0007 | -0.0001 | 0.0036 | 0.083 NMR | 0.77% | 40.7% | — |
| 293 | agg | JOS_KAGGLE_SUNSHINE | 0.0208 | 0.0104 | 0.0107 | 0.0018 | 0.0023 | 0.0081 | 0.069 NMR | 0.98% | 34.7% | — |
| 1114 | agg | JOS_KAGGLE_SHATTER | 0.0136 | 0.0034 | 0.0084 | 0.0002 | 0.0009 | 0.0025 | 0.073 NMR | 0.02% | 24.0% | — |
| 1409 | agg | JOS_KAGGLE_MEDIUM_FN | 0.0086 | 0.0028 | 0.0066 | -0.0007 | -0.0005 | 0.0012 | 0.069 NMR | 0.39% | 21.2% | — |
| 2725 | agg | JOS_KAGGLE_MEDIUM_TE | 0.0093 | 0.0035 | 0.0066 | -0.0016 | -0.0016 | 0.0024 | 0.063 NMR | 0.77% | 12.7% | — |

JOS_KAGGLE_SHATTER

Woke: Jun 19, 2023 | [Account summary »](#)

 - ☒ Public Kaggle notebook
 - ☒ Getting started tutorial #1 - Build your 1st model - modified to run in Kaggle with better results

Performance

	1Y Avg	Rank
MMC	0.0002	8251
CORR20V2	0.0084	1627
CORJ60	0.0136	2245
FNCV3	0.0034	5679
BMC	0.0009	6227
CORT20	0.0025	5903

Current Model Correlations

CWMM	0.7034
MCWNM	1
APCWNM	0.3993

Stake

0.0725 NMR

1 Day Return	▲ 0.02%
3 Month Return	▲ 24.0%
1 Year Return	—
All Time Return	▲ 45.0%
Multipliers	0.5xCORR20v2 2xMMC
Meta Model Control	0.0000%

JOS KAGGLE HELLO

HELLO NUMERAI AUTOMATED (KAGGLE.COM)

Typically consists from 3 main parts:

- Original example code slightly modified for Kaggle
 - Using Kaggle public dataset
 - Modifying Kaggle environment
- Model upload with NumerAPI
- Improving model performance usually by training on medium feature set

Modifications:

- [Hello Numerai - NumerAPI submissions \(kaggle.com\)](#) – submitting predictions via NumerAPI
- [Hello Numerai automated on GPU \(kaggle.com\)](#) – training on P100 GPU

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▲ 3

Edit

Hello Numerai automated

Python · [numerai - V4.3 Midnight](#), [Hello Numerai automated](#)

Notebook Input Output Logs Comments (0) Settings

Run1260.6sVersion 27 of 27

Add Tags



Hello [Numerai](#), this is Kaggle!

Numerai is "the hardest data science tournament in the world". Despite its challenging nature it is very easy to join and participate. This notebook is probably the quickest way for newcomer to complete the simple model training and submit predictions using newly [announced model upload capability](#). Uploaded models are automatically executed against current out-of-sample "live" data to submit tournament predictions.

It is modified version of Google Colab notebook from [numerai's example scripts git repository](#) to run in Kaggle environment. Main modifications are in three areas:

- There is weekly updated Kaggle dataset with full [latest version V4.3 dataset](#) and also another dataset with [V4.2, V4 and V4.1 Numerai data](#), so that you do not need to download it every time. The v4.2 dataset is attached to this notebook as input

Table of Contents >

- Hello Numerai, this is Kaggle!
- 1. Dataset
- 2. Modeling
- 3. Submissions
- That's it. Now upload your model!
- Want better performance?
- Better model is ready for upload!

JOS KAGGLE SUNSHINE

NUMERAI EXAMPLE MODEL SUNSHINE (KAGGLE.COM)

- Using old Numerai `utils` library for imports
- Both ensemble and neutralization
- Discussion of submission options
- Concluding remarks on various performance enhancement techniques

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5 Edit

Numerai Example Model Sunshine

Python · [numerai data v4.2 Rain](#), [utils](#), [Numerai Example Model Sunshine](#)

Notebook Input Output Logs Comments (0) Settings

Run
9262.4s

Version 17 of 17

Add Tags

 **NUMERAI**

Numerai Example Model Sunshine

[Numerai is "the hardest data science tournament in the world"](#). Despite its challenging nature it is very easy to join and participate. To help you get started with building your own models, the Numerai team has built a [number of example models](#). This notebook is Kaggle version of [Sunshine example script](#) (working with V4.2 Rain dataset - so it should be renamed to Rain Example ;-)) and exploring some intermediate concepts like ensembling and normalizing. If you are complete Numerai novice, you could check also these two Kaggle notebooks for quickstart:

- [Hello Numerai - NumerAPI submissions](#) - the quickest way for newcomer to complete the simple model training and submit predictions "manually" by running this notebook
- [Hello Numerai automated](#) - version of previous notebook prepared for upload to Numerai platform for fully automated daily submission of your predictions. This model contains also simple "tweak" to improve diagnostic performance and hopefully tournament results.

Table of Contents >

- Numerai Example Model Sunshine
- Install and import libraries
- Load the datasets
- Copy trained models
- Train models
- Make ansamble model and its featu...
- Prepare predictions for diagnostics...
- Calculate validation metrics
- Prepare model upload for automate...
- Upload trained model via NumerAPI
- Concluding remarks
- The End

OTHER NOTEBOOK RESOURCES

Hyperparameter search:

- [Numerai LightGBM Hyperparameter Search with Optuna \(kaggle.com\)](#)
- [Numerai XGBoost Hyperparameter Search with Optuna \(kaggle.com\)](#)
- [Numerai CatBoost Hyperparameter Search with Optuna \(kaggle.com\)](#)

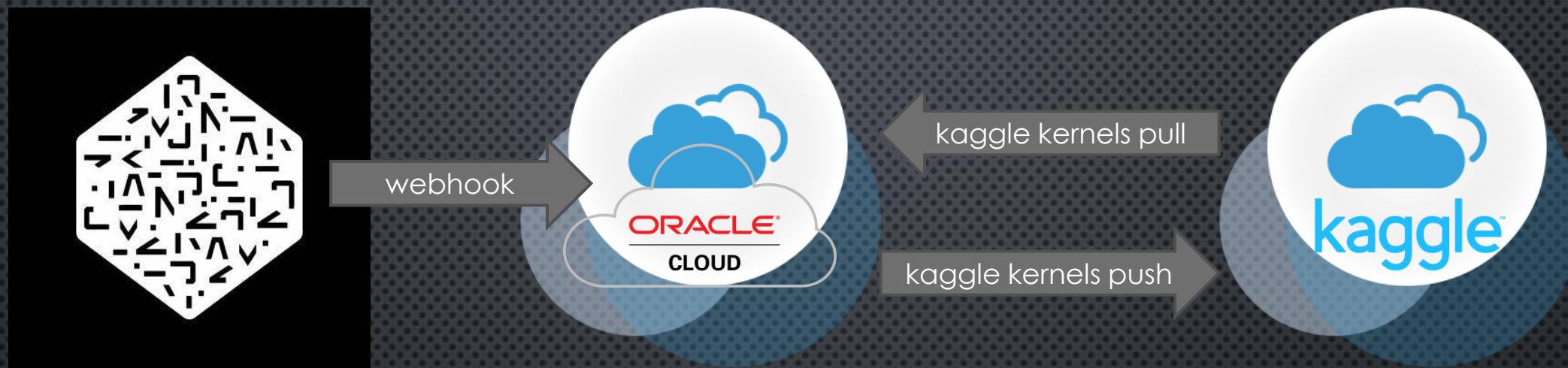
KATSU1110 – Numerai

- [\[numerai\] MLP with KerasTuner Starter \(kaggle.com\)](#)
- [\[Numerai\] NN baseline with new massive data \(kaggle.com\)](#)
- [\[numerai\] factor analysis \(kaggle.com\)](#)
- [\[numerai\] PCA magic \(kaggle.com\)](#)
- [\[NumeraiSignals\] Starter for Beginners \(kaggle.com\)](#)

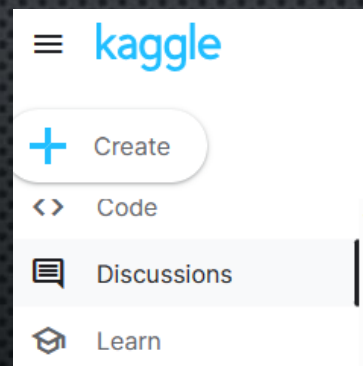
Carlo Lepelaars | Master | Kaggle of CrowdCent

- [How to get started with Numerai \(kaggle.com\)](#)
- [Getting Started with Numerai Signals \(kaggle.com\)](#)
- [Numerai inference pipelines with NumerBlox \(kaggle.com\)](#)

TRIGGERING RUN ON KAGGLE



- ORACLE CLOUD, „ALWAYS FREE“, UBUNTU VM, 1 CPU WITH BOOT DISK
- PYTHON
- KAGGLE API CLI
- FLASK WEBSERVER

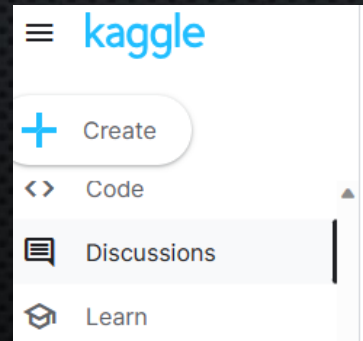


Search



JOSEF ŠVENDA · POSTED 3 MONTHS AGO IN [QUESTIONS & ANSWERS](#)

Triggering notebook execution by webhook with Flask and Kaggle API



Search

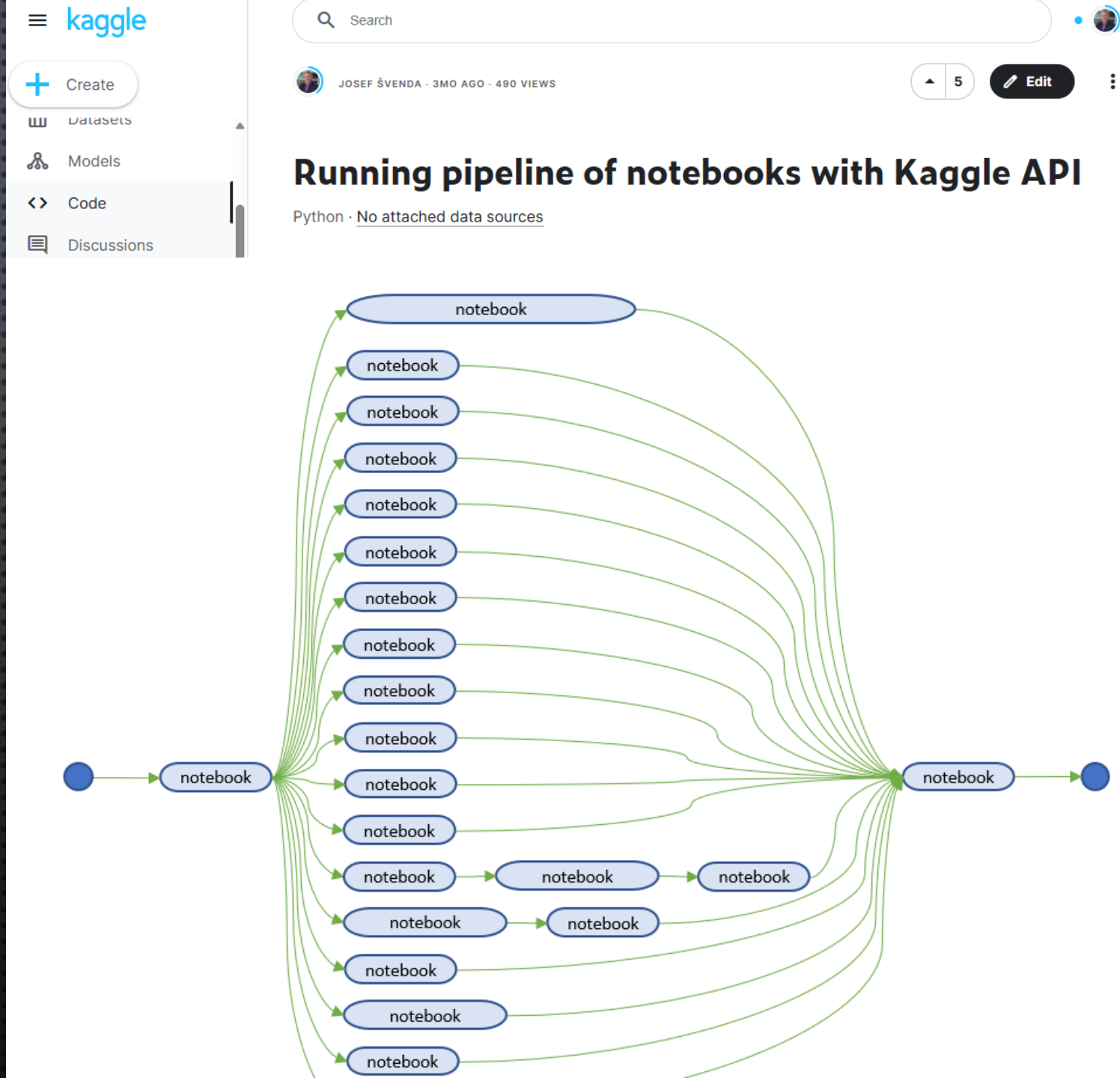


JOSEF ŠVENDA · POSTED 7 MONTHS AGO IN [PRODUCT FEEDBACK](#)

Scheduling notebook execution with cron and Kaggle API

KAGGLE NOTEBOOK ORCHESTRATOR

- RUNNING OTHER NOTEBOOKS
 - SEQUENTIALLY
 - PARALLEL
- MUST FINISH IN 12 HOURS
- KAGGLE API CLI
- CONTROLLING PYTHON DICT



BEST THINGS IN LIFE ARE FREE!!!

... BUT GOOD FINANCIAL REWARD IS ALWAYS APPRECIATED ;-)